

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1928

Gwf Tracy 138 kV
December 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1928. The Interconnection Customer proposes a 144.2 MW solar photovoltaic facility interconnecting at the Gwf Tracy 138 kV substation in the Pennsylvania area. The request was received on November 08, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 650 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,422,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of RHRERSTOWN - GWF TRACY 138.0 kV, the IRVING BELT LINE - ROHRERSTOWN 138.0 kV L loads to 132.1% of its 255 MVA rating. Project DFAX: 37.6%.

GWF TRACY - ROHRERSTOWN 138.0 kV: 115.9% loading (505 MVA) following the outage of the ROCKY MTN HYDROELECTRIC PLANT - DONERVILLE 138.0 kV; DFAX = 74.6%.

For the loss of GWF TRACY - ROHRERSTOWN 138.0 kV, the RD - IRVING BELT LINE 138.0 kV loads to 111.9% of its 333 MVA rating. Project DFAX: 37.6%.

For the loss of PENNZEIR 345-138 kV XFMR, the W CANON - FULLER TAP 138.0 kV line loads to 110.2% of its 174 MVA rating. Project DFAX: 7.7%.

For the loss of PENN ZEIR 345-138 kV Transformer, the ROHRERSTOWN - LANCASTER AIR PRODUCTS 138.0 kV loads to 106.1% of its 238 MVA rating. Project DFAX: 7.7%.

MT JOY - LANCASTER AIR PRODUCTS 138.0 kV Line: 105.0% loading (230 MVA) following the outage of the PENN ZEIR 345 kV/138 kV Autotransformer; DFAX = 7.7%.

The ELK RIVER - FLANNIGAN 345.0 kV is overloaded to 104.6% of its 1,065 MVA emergency rating for the loss of the TOMBSTOE - BIXBY HILL 500.0 kV. The project contributes 10.5% DFAX (15.2 MW) to this overload.

For the loss of PENN ZEIR 345/138kV Auto, the W CANON SUBSTATION - PORT WING 138.0 kV L loads to 104.2% of its 197 MVA rating. Project DFAX: 7.7%.

PORT WING - MT JOY 138.0 kV: 102.2% loading (212 MVA) following the outage of the PENN ZEIR 345/138kV TX; DFAX = 7.7%.

MERCURY - ROHRERSTOWN 345.0 kV: 101.0% loading (1,099 MVA) following the outage of the IRONWOOD LLC - FAYETTE 345.0 kV; DFAX = 23.0%.

The KINGSTON - IPL FIT MONROVIA I LLC 138.0 kV line is overloaded to 100.8% of its 334 MVA emergency rating for the loss of the MERCURY SUBSTATION - ROHRERSTOWN 345.0 kV. The project contributes 12.5% DFAX (18.1 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
IRVING BELT LINE - ROHRERSTOWN 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,774,000	12% (\$321,000)
GWF TRACY - ROHRERTOWN 138.0 kV	Rebuild line with higher-capacity conductor	\$1,764,000	18% (\$316,000)
RD - IRVING BELT LINE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,660,000	12% (\$192,000)
W CANON - FULLER TAP 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,671,000	37% (\$2,091,000)
ROHRERSTOWN - LANCASTER AIR PRODUCTS 138.0 kV	Reconductor line	\$792,000	37% (\$292,000)
MT JOY SUBSTATION - LANCASTER AIR PRODUCTS 138.0 kV L	Reconductor line	\$1,952,000	37% (\$720,000)
ELK RIV - FLANNIGAN 345.0 kV	Reconductor line	\$11,581,000	3% (\$314,000)
W CANON SUBSTATION - PORT WING 138.0 kV L	Reconductor line	\$2,143,000	37% (\$790,000)
PORT WING - MT JOY 138.0 kV	Reconductor line	\$1,408,000	37% (\$519,000)
MERCURY - ROHRERSTOWN 345.0 kV	Reconductor line	\$10,241,000	6% (\$578,000)
KINGSTON - IPL FIT MONROVIA I LLC 138.0 kV line	Reconductor line	\$4,151,000	7% (\$289,000)
	Total allocated to project		\$6,422,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.